

Color Guided Lab

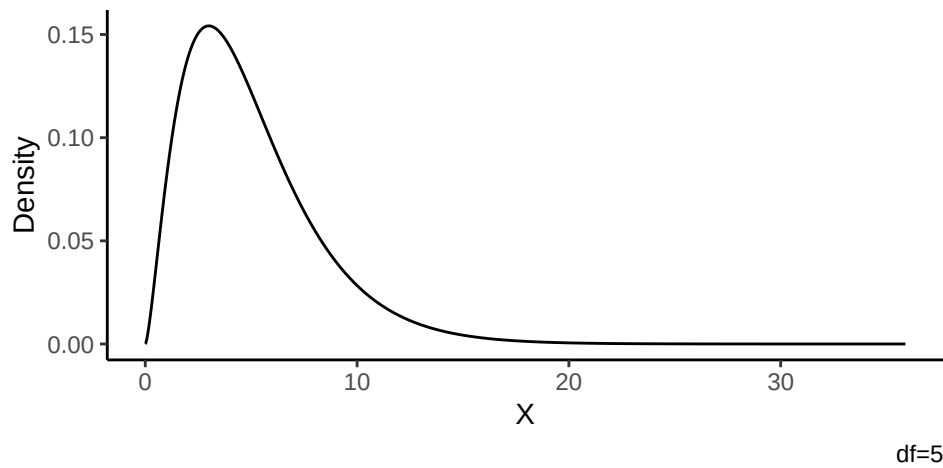
Robert Thomas Kohn

9/22/25

Start Graphing!

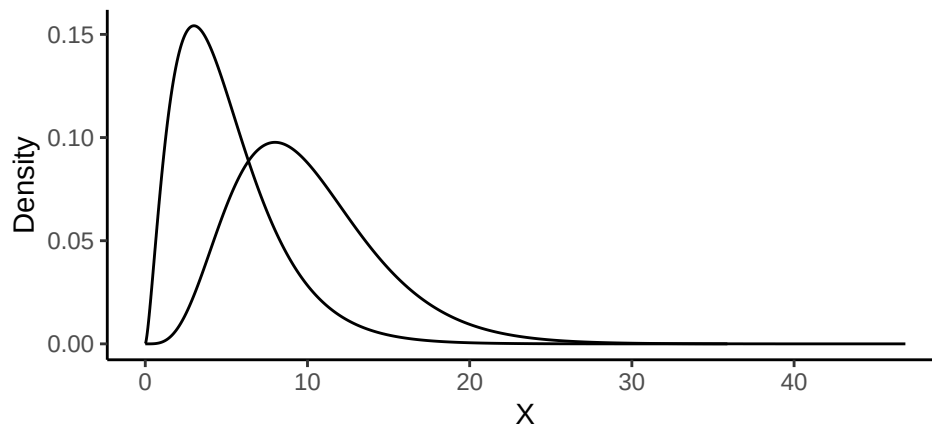
```
gf_dist("chisq", df = 5)%>%gf_labs(x = "X", y = "Density", title="Showcasing Chi Squared Relationships")
```

Showcasing Chi Squared Relationships



```
gf_dist("chisq", df = 5)%>%gf_dist("chisq", df = 10)%>%gf_labs(x = "X", y = "Density", title="Showcasing")
```

Showcasing Chi Squared Relationships

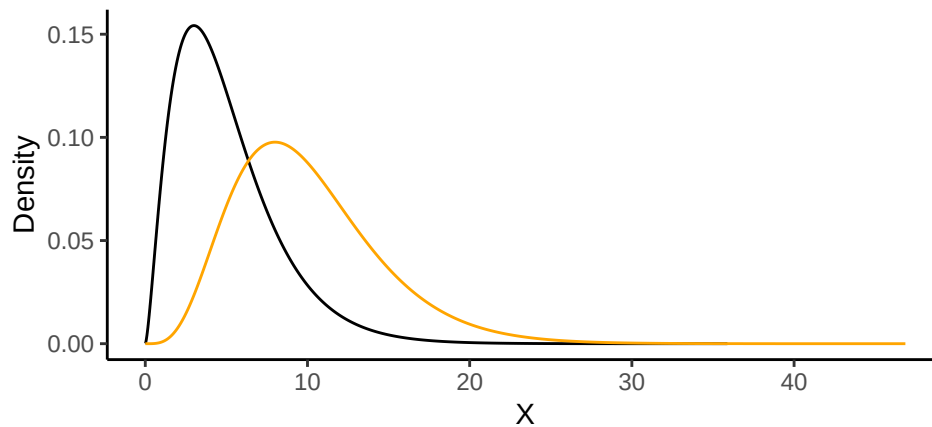


df=5 and df=10

```
gf_dist("chisq", df = 5)%>%gf_dist("chisq", df = 10,color="orange")%>%gf_labs(x = "X", y = "Density", t
```

Showcasing Chi Squared Relationships

df=5 is Black, df=10 is Orange

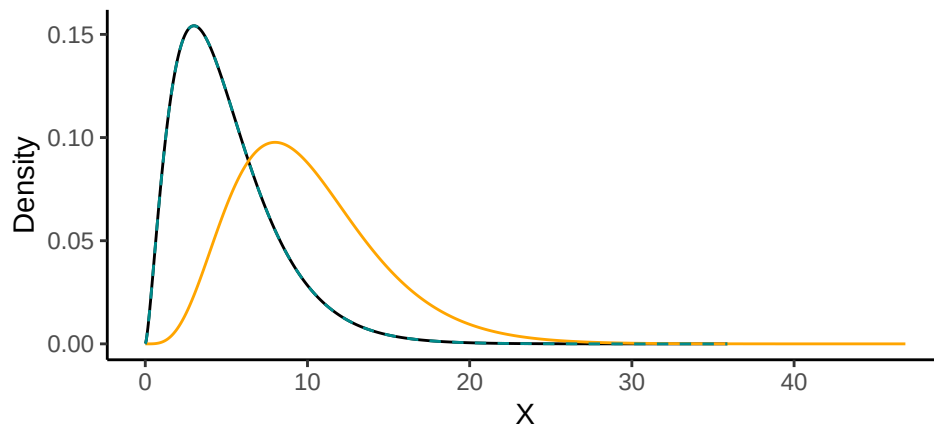


df=5 and df=10

```
gf_dist("chisq", df = 5)%>%gf_dist("chisq", df = 10,color="orange")%>%gf_dist("chisq", df = 5,color="da
```

Showcasing Chi Squared Relationships

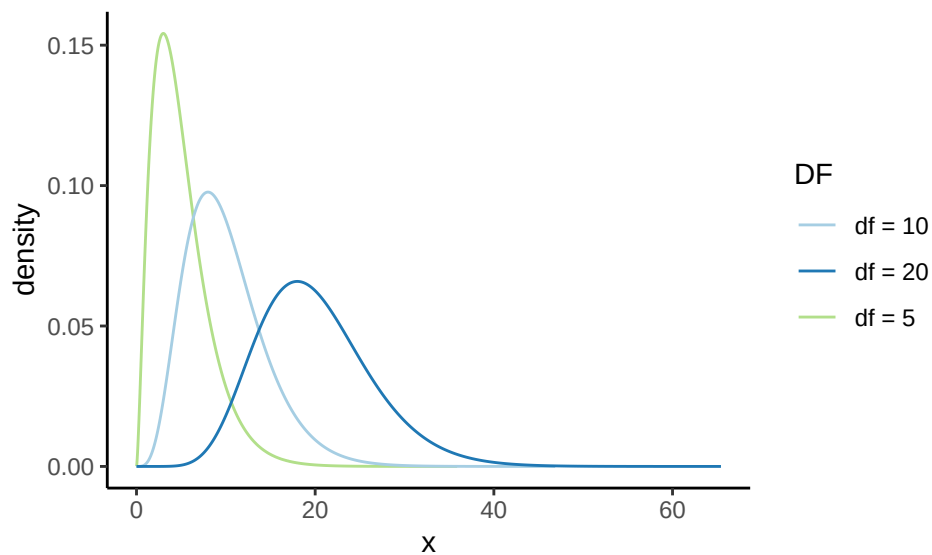
df=5 is Black with Dark Cyan Dashes, df=10 is Orange



2 df=5 and df=10

#Alternative Option

```
gf_dist("chisq", df = 5, color = ~ "df = 5")%>%gf_dist("chisq", df = 10, color = ~ "df = 10")%>%gf_dist
```

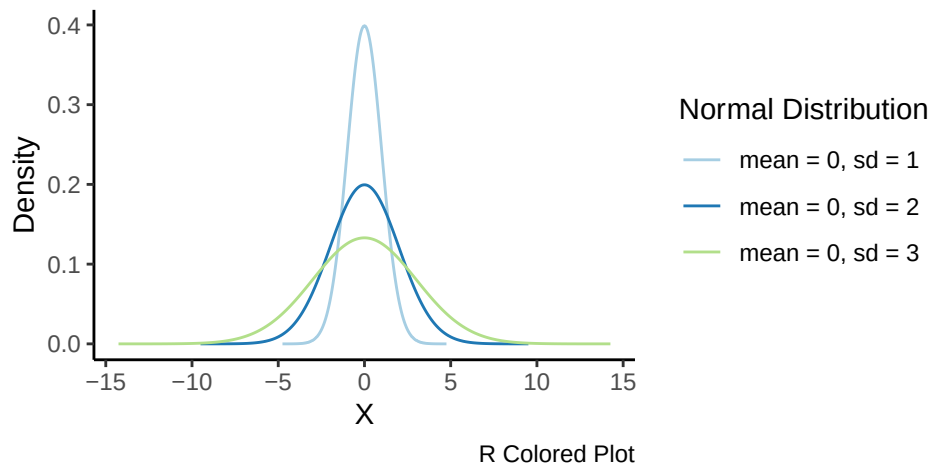


Your Turn!! - Normal Distribution

```
gf_dist("norm", mean = 0, sd = 1, color = ~ "mean = 0, sd = 1") %>% gf_dist("norm", mean = 0, sd = 2, color = ~ "mean = 0, sd = 2")
```

Normal Distribution Relationships as a Result of SD

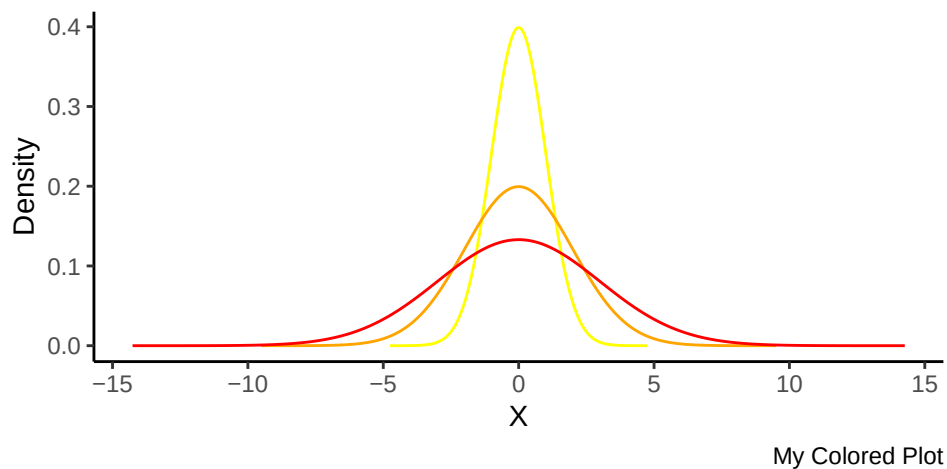
Cyan=1, Blue=2, Green=3



```
gf_dist("norm", mean = 0, sd = 1,color="yellow")>%gf_dist("norm", mean = 0, sd = 2,color="orange")>%g
```

Normal Distribution Relationships as a Result of SD

Yellow=1SD, Orange=2SD, Red=3SD



Conditional Coloring

```
require(MASS)
```

```
## Loading required package: MASS
```

```
##
```

```
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
## select
```

```
library(ggformula)
```

```
data(Cars93)
```

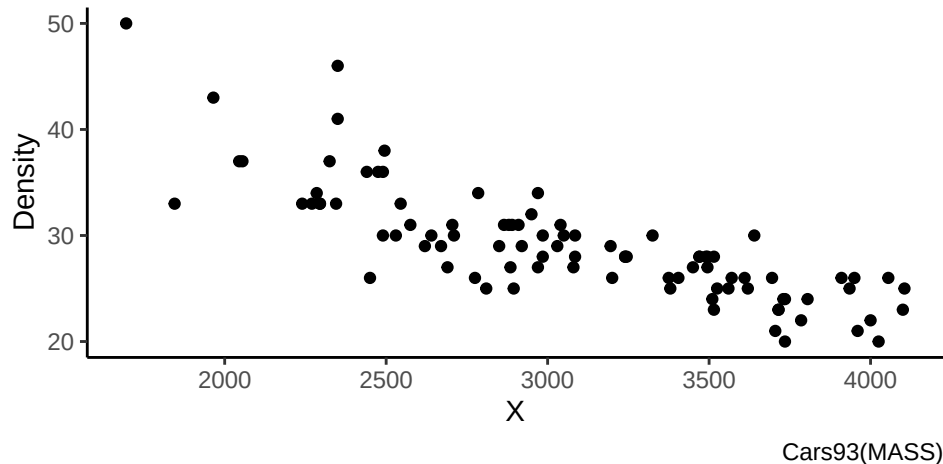
```
head(Cars93)
```

```
##      Manufacturer  Model   Type Min.Price Price Max.Price MPG.city MPG.highway
## 1      Acura  Integra   Small    12.9  15.9    18.8     25        31
## 2      Acura  Legend Midsize    29.2  33.9    38.7     18        25
## 3      Audi    90  Compact    25.9  29.1    32.3     20        26
## 4      Audi   100 Midsize    30.8  37.7    44.6     19        26
## 5      BMW    535i Midsize    23.7  30.0    36.2     22        30
## 6     Buick Century Midsize    14.2  15.7    17.3     22        31
##
##           AirBags DriveTrain Cylinders EngineSize Horsepower  RPM
## 1              None      Front         4         1.8        140 6300
## 2 Driver & Passenger      Front         6         3.2        200 5500
## 3      Driver only      Front         6         2.8        172 5500
## 4 Driver & Passenger      Front         6         2.8        172 5500
## 5      Driver only      Rear          4         3.5        208 5700
## 6      Driver only      Front          4         2.2        110 5200
##
## Rev.per.mile Man.trans.avail Fuel.tank.capacity Passengers Length Wheelbase
## 1      2890              Yes          13.2           5     177     102
## 2      2335              Yes          18.0           5     195     115
## 3      2280              Yes          16.9           5     180     102
## 4      2535              Yes          21.1           6     193     106
## 5      2545              Yes          21.1           4     186     109
## 6      2565              No          16.4           6     189     105
##
## Width Turn.circle Rear.seat.room Luggage.room Weight  Origin      Make
## 1    68          37          26.5          11  2705 non-USA  Acura Integra
## 2    71          38          30.0          15  3560 non-USA  Acura Legend
## 3    67          37          28.0          14  3375 non-USA    Audi 90
## 4    70          37          31.0          17  3405 non-USA    Audi 100
## 5    69          39          27.0          13  3640 non-USA    BMW 535i
## 6    69          41          28.0          16  2880   USA Buick Century
```

```
gf_point(MPG.highway ~ Weight , data = Cars93 )>%gf_labs(x = "X", y = "Density", title="The Heavier the
```

The Heavier the Car, the Less MPG on the Highway

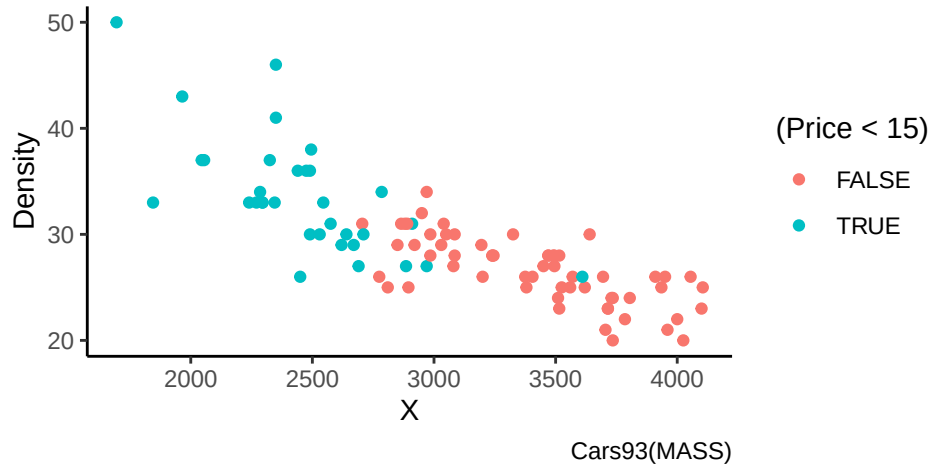
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Price < 15) )>%gf_labs(x = "X", y = "Density")
```

The Heavier the Car, the Less MPG on the Highway

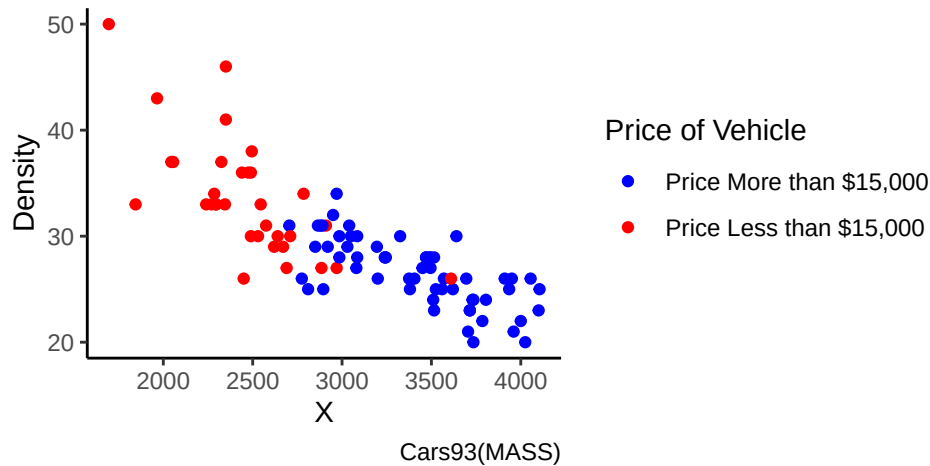
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Price < 15) )>%gf_refine(scale_color_manual(v
```

The Heavier the Car, the Less MPG on the Highway

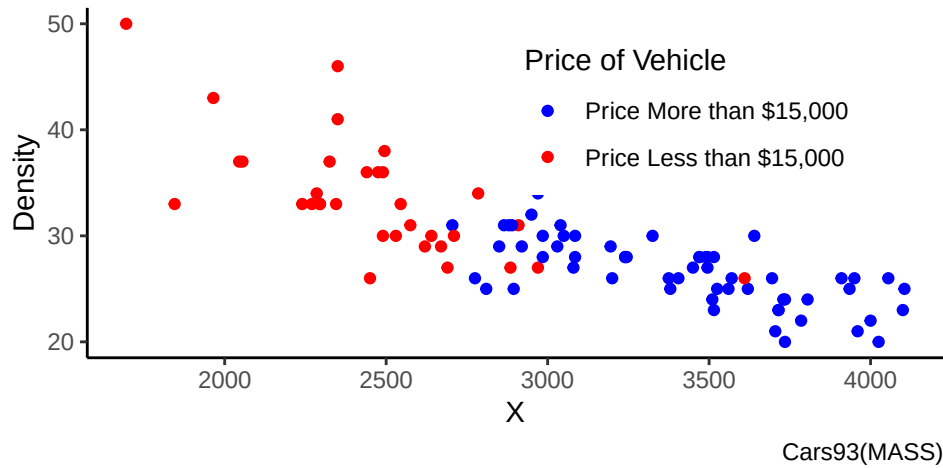
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Price < 15) )>%gf_refine(scale_color_manual(v
```

The Heavier the Car, the Less MPG on the Highway

Scatterplot of Weight vs MPG on the Highway

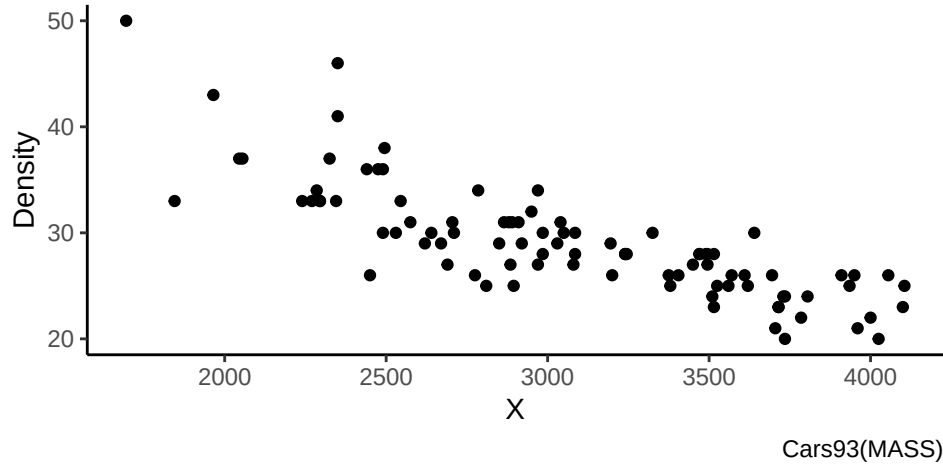


Color based on Categorical Condition

```
gf_point(MPG.highway ~ Weight , data = Cars93 )>%gf_labs(x = "X", y = "Density", title="The Heavier th
```

The Heavier the Car, the Less MPG on the Highway

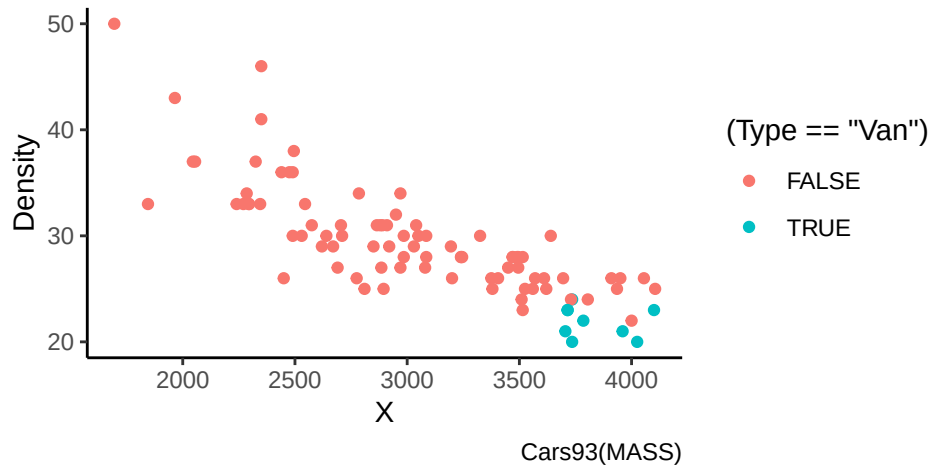
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Type=="Van") )>%gf_labs(x = "X", y = "Density
```

The Heavier the Car, the Less MPG on the Highway

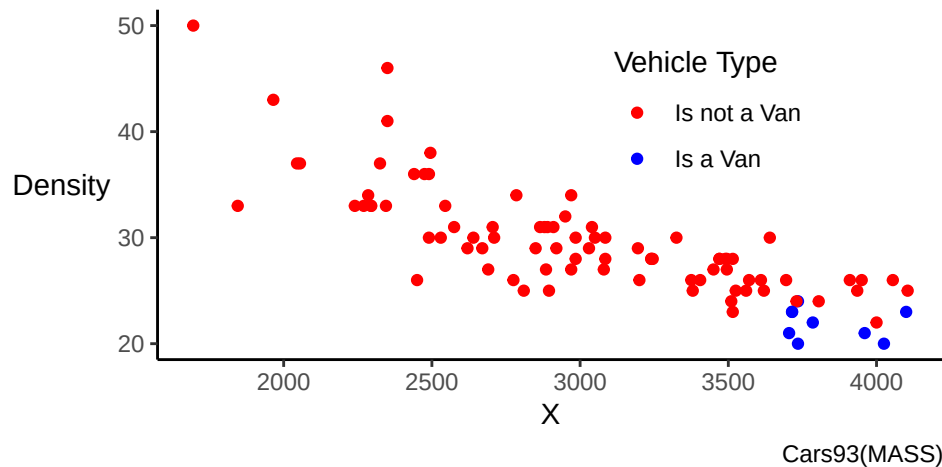
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Type=="Van")) %>% gf_refine(scale_color_manual(
```

The Heavier the Car, the Less MPG on the Highway

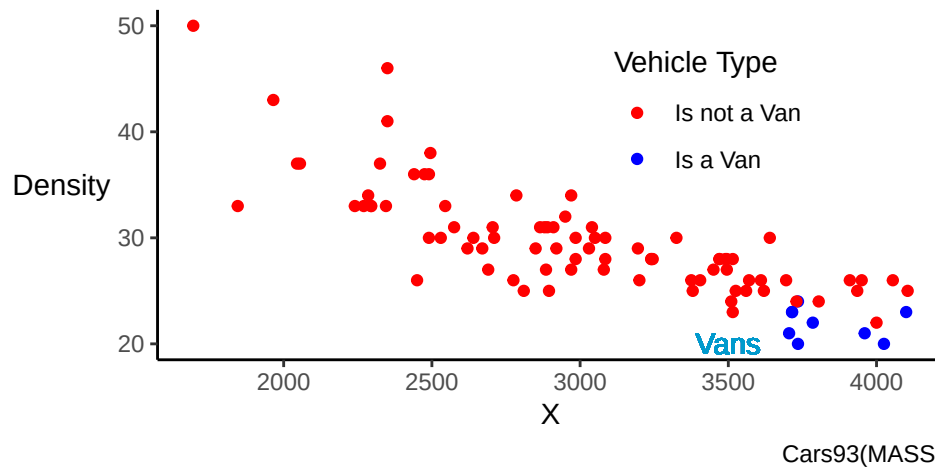
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Type=="Van")) %>% gf_refine(scale_color_manual(
```

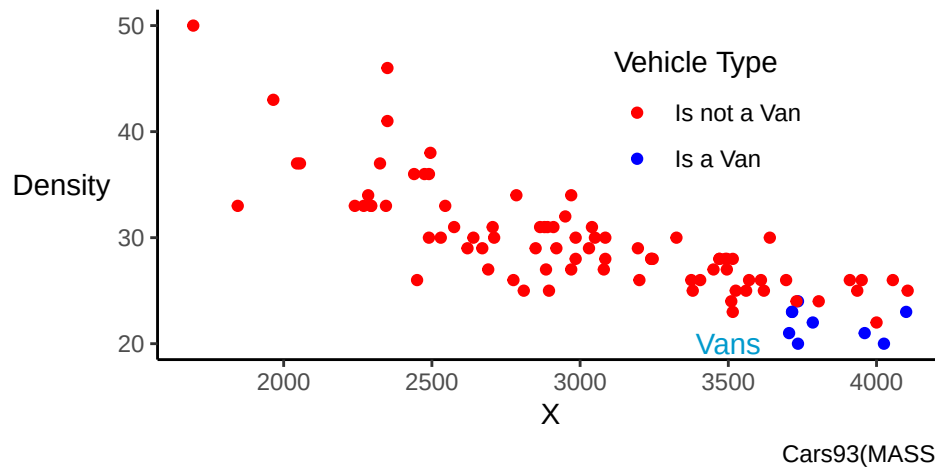
```
## Warning: All aesthetics have length 1, but the data has 93 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

The Heavier the Car, the Less MPG on the Highway
Scatterplot of Weight vs MPG on the Highway



```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Type=="Van")) ) %>%gf_refine(annotate(geom = "t
```

The Heavier the Car, the Less MPG on the Highway
Scatterplot of Weight vs MPG on the Highway



Categorical Colors

```
library(palmerpenguins)
```

```
## Warning: package 'palmerpenguins' was built under R version 4.4.3
```

```
data(penguins)
```

```
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>
## 1 Adelie Torgersen     39.1          18.7           181          3750
## 2 Adelie Torgersen     39.5          17.4           186          3800
```

```
## 3 Adelie Torgersen      40.3      18      195      3250
## 4 Adelie Torgersen      NA       NA       NA       NA
## 5 Adelie Torgersen      36.7     19.3     193     3450
## 6 Adelie Torgersen      39.3     20.6     190     3650
## # i 2 more variables: sex <fct>, year <int>
```

```
require(viridis)
```

```
## Loading required package: viridis
```

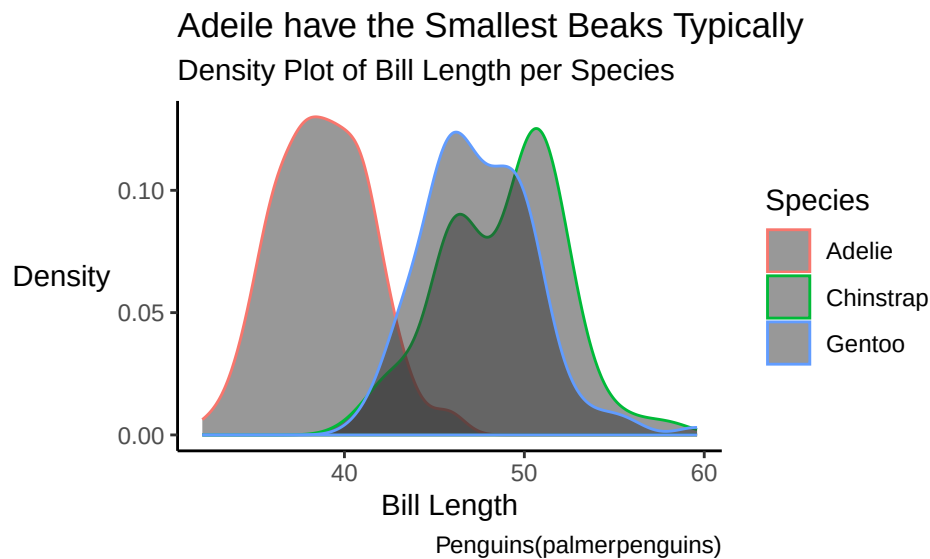
```
## Warning: package 'viridis' was built under R version 4.4.3
```

```
## Loading required package: viridisLite
```

```
gf_density( ~ bill_length_mm , data = penguins , color = ~ species )>%gf_labs(x = "Bill Length", y = "Density")
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## (`stat_density()`).
```



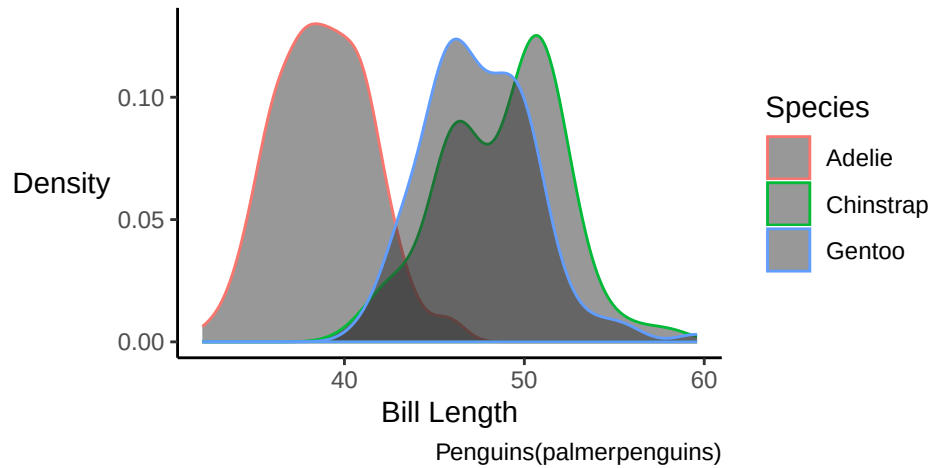
```
gf_density( ~ bill_length_mm , data = penguins , color = ~ species )>%gf_refine( scale_fill_manual( values = c("#F08080", "#4DC0B5", "#6495ED") ) )
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## (`stat_density()`).
```

Adeile have the Smallest Beaks Typically

Density Plot of Bill Length per Species



```
gf_density( ~ bill_length_mm , data = penguins , color = ~ species )>%gf_refine( scale_fill_brewer( pa
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

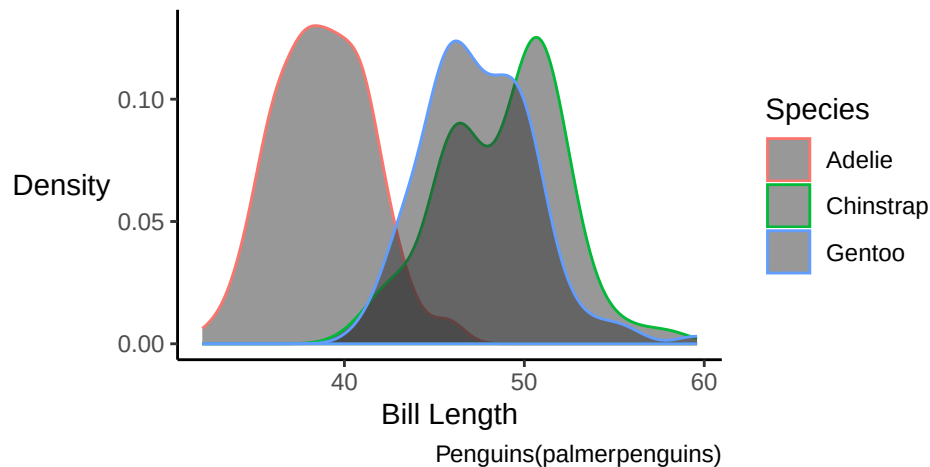
```
## (`stat_density()`).
```

```
## Warning in structure(in_domain, pos = match(in_domain, breaks)): Calling 'structure(NULL, *)' is dep
```

```
## Consider 'structure(list(), *)' instead.
```

Adeile have the Smallest Beaks Typically

Density Plot of Bill Length per Species



```
gf_density( ~ bill_length_mm , data = penguins , color = ~ species )>%gf_refine( scale_fill_viridis( d
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

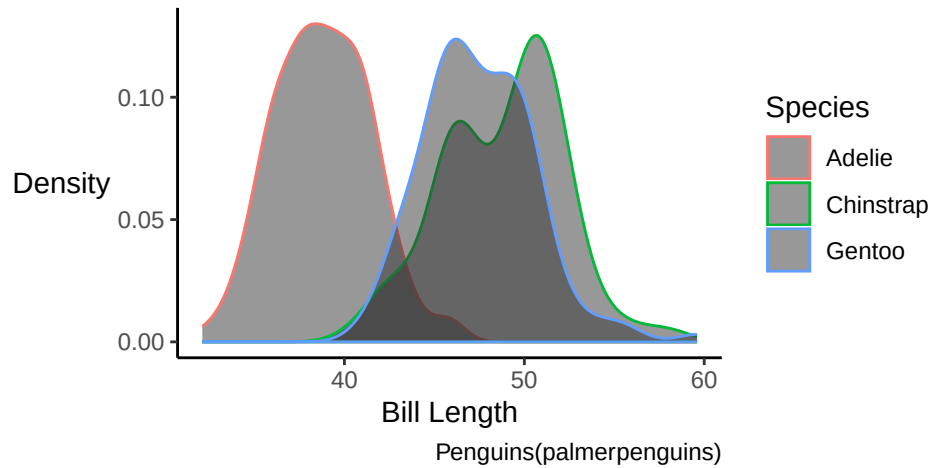
```
## (`stat_density()`).
```

```
## Calling 'structure(NULL, *)' is deprecated, as NULL cannot have attributes.
```

```
## Consider 'structure(list(), *)' instead.
```

Adeile have the Smallest Beaks Typically

Density Plot of Bill Length per Species



```
gf_histogram(~bill_length_mm, data = penguins, fill=~species, alpha = 0.5, position = position_identity)
```

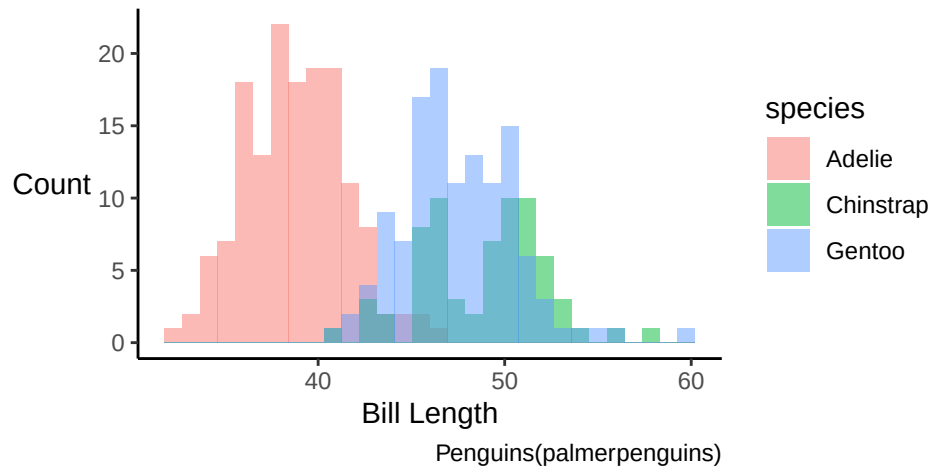
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## (`stat_bin()`).
```

Adeile have the Smallest Beaks Typically

Histogram of Bill Length per Species



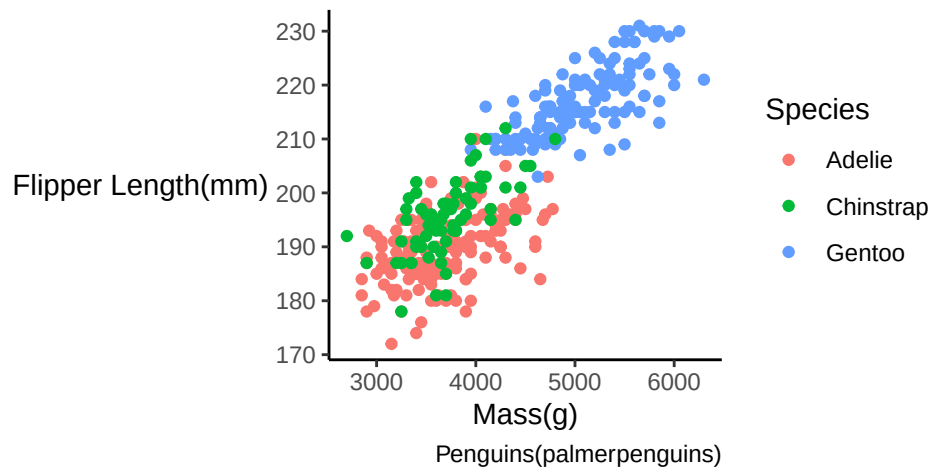
Highlight Using Discrete Color

```
gf_point(flipper_length_mm ~ body_mass_g, data = penguins, color = ~(species))%>%gf_labs(x = "Mass(g)")
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

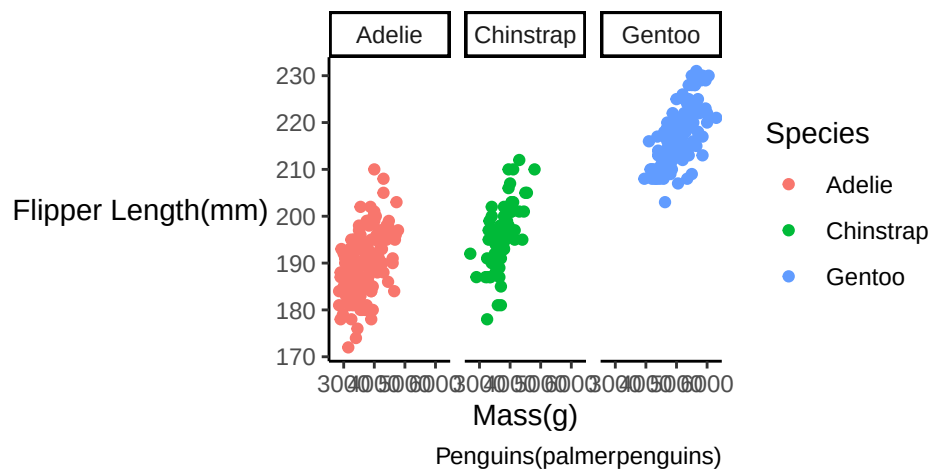
Gentoo Penguins have the longest flipper
Scatterplot of Weight vs Flipper Length when cor



```
gf_point(flipper_length_mm ~ body_mass_g , data = penguins, color = ~species)%>%gf_facet_wrap(~species)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

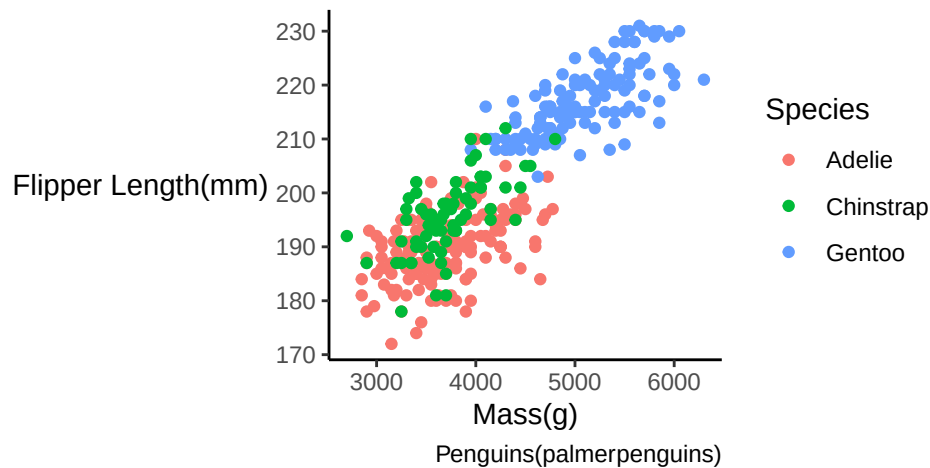
Gentoo Penguins have the longest flipper
Scatterplot of Weight vs Flipper Length separate



```
gf_point(flipper_length_mm~body_mass_g, data=transform(penguins, species = NULL), color = "gray85", alp
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

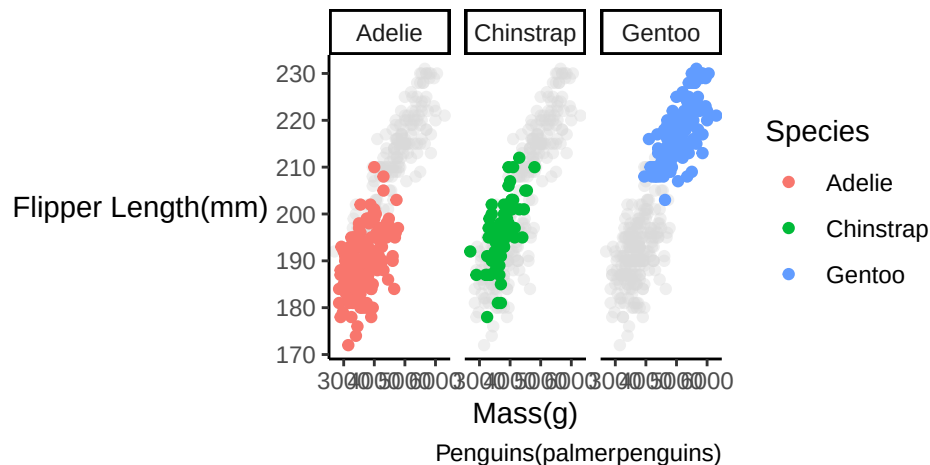
Gentoo Penguins have the longest flipper
Scatterplot of Weight vs Flipper Length when cor



```
gf_point(flipper_length_mm~body_mass_g, data=transform(penguins, species = NULL), color = "gray85", alp
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Gentoo Penguins have the longest flipper
Scatterplot of Weight vs Flipper Length seperate



Continuous Color

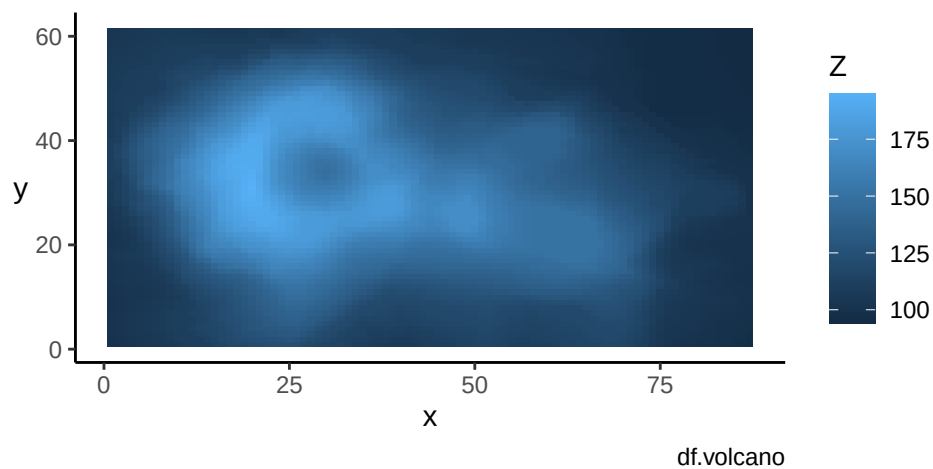
```
require(tidyr)
```

```
## Loading required package: tidyr
##
## Attaching package: 'tidyr'
## The following objects are masked from 'package:Matrix':
```

```
##
##      expand, pack, unpack
df.volcano = as.data.frame(volcano) %>%
mutate(X = seq(from = 1, to = 87, by=1)) %>%
pivot_longer(cols = starts_with("V"), names_to = "Y", names_prefix = "V", values_to = "Z") %>%
mutate(Y = as.numeric(Y))

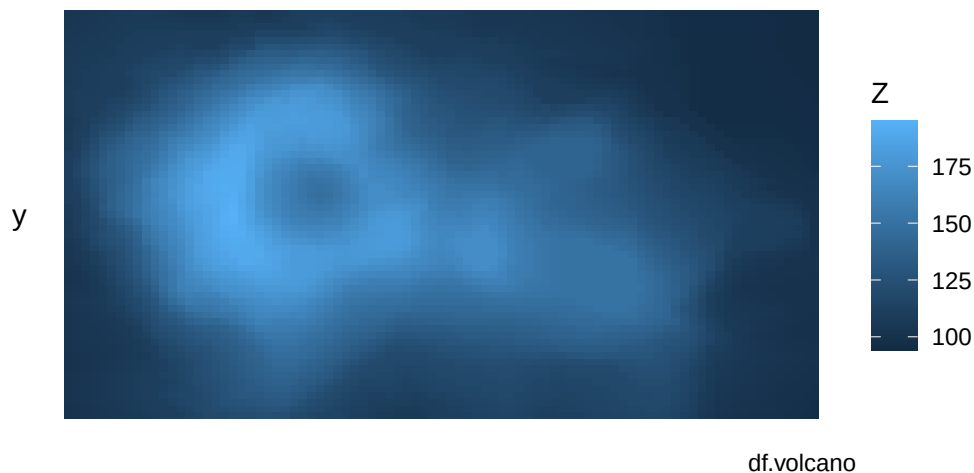
gf_raster(Z~X+Y, data = df.volcano)%>%gf_labs(x = "x", y = "y", title="Volcano Plot" , subtitle = "" ,
```

Volcano Plot



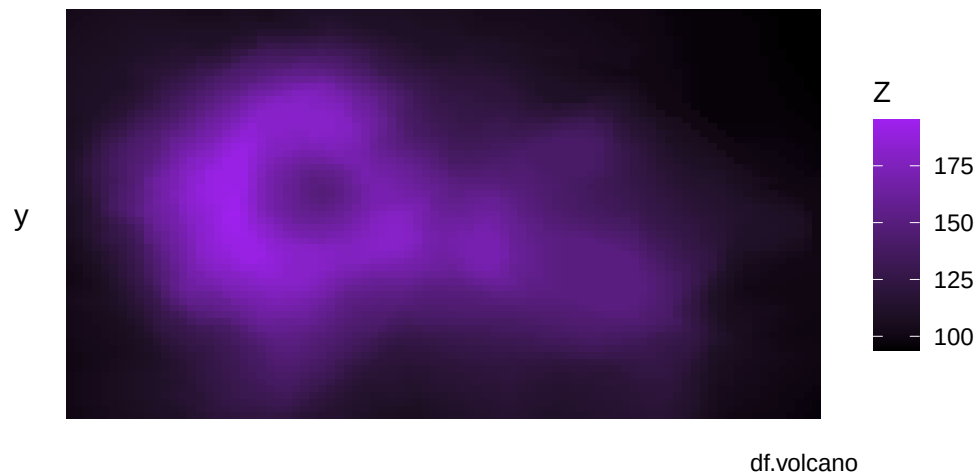
```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_labs(x = "x", y = "y", title="Volcano
```

Volcano Plot



```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void()) %>%gf_refine( scale_fill_gradient( low = "
```

Volcano Plot



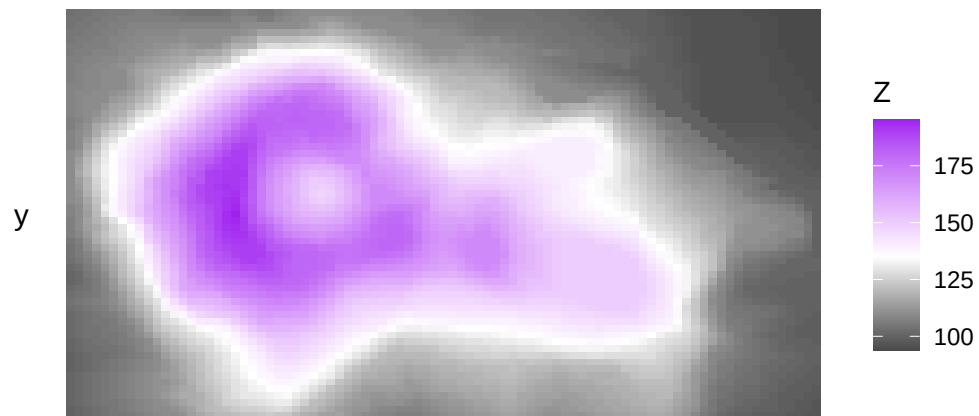
```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine(scale_fill_gradient2(low = "bl
```

Volcano Plot



```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine(scale_fill_gradient2(low = "bl
```

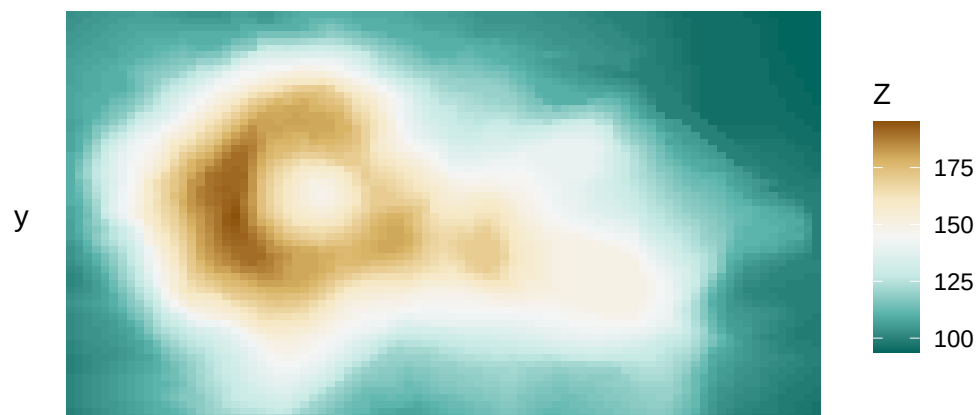
Volcano Plot



df.volcano

```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine( scale_fill_distiller( palette =
```

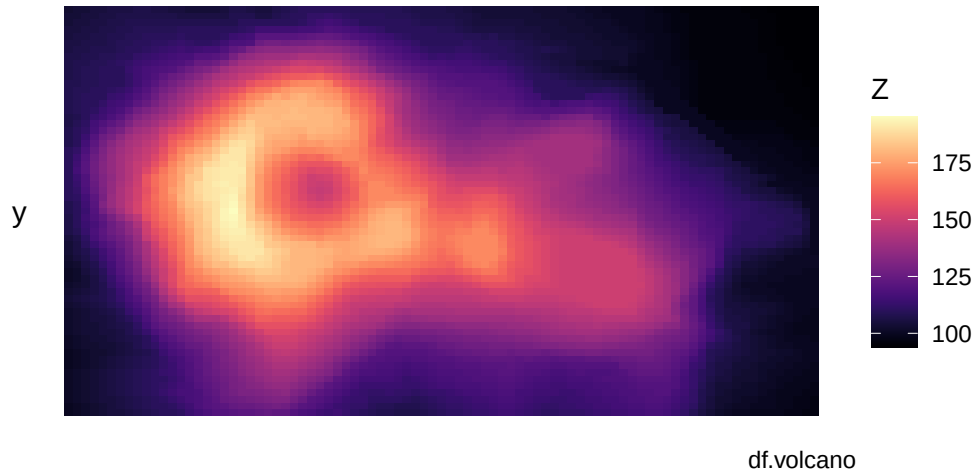
Volcano Plot



df.volcano

```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine( scale_fill_viridis( option =
```

Volcano Plot



```
library(readr)
temps_months <- read_csv("https://wilkelab.org/SDS375/datasets/tempnormals.csv") %>%
  group_by(location, month_name) %>%
  summarize(mean = mean(temperature)) %>%
  mutate(
    month = factor(
      month_name,
      levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
    ),
    location = factor(
      location, levels = c("Death Valley", "Houston", "San Diego", "Chicago")
    )
  ) %>%
  dplyr::select(-month_name)

## Rows: 1464 Columns: 8

## -- Column specification -----
## Delimiter: ","
## chr  (4): location, station_id, month_name, month
## dbl  (3): temperature, day, day_of_year
## date (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has grouped output by 'location'. You can override using the `.groups` argument.

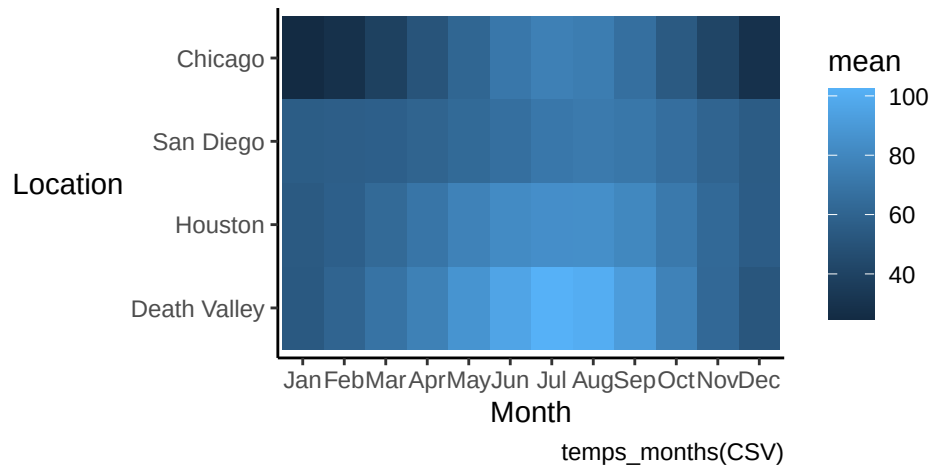
head(temps_months)

## # A tibble: 6 x 3
## # Groups:   location [1]
##   location mean month
##   <fct>     <dbl> <fct>
## 1 Chicago  50.4 Apr
## 2 Chicago  74.1 Aug
## 3 Chicago  29   Dec
```

```
## 4 Chicago    28.9 Feb
## 5 Chicago    24.8 Jan
## 6 Chicago    75.8 Jul
```

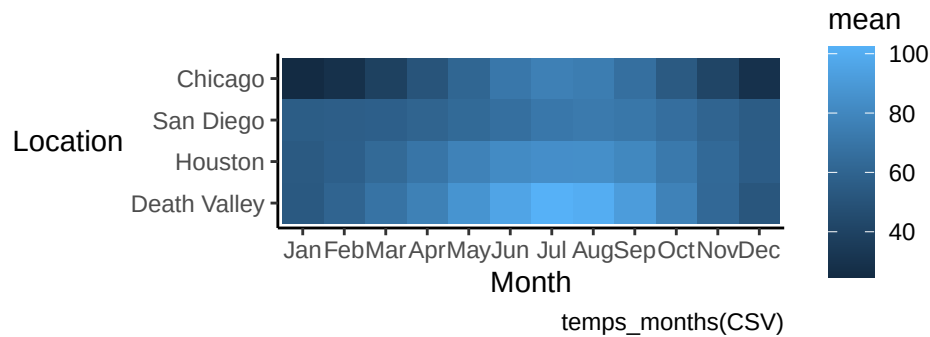
```
gf_tile( location ~ month , data = temps_months , fill = ~ mean) %>% gf_labs(x = "Month", y = "Location")
```

Chicago is abnormally cool compared to other locations

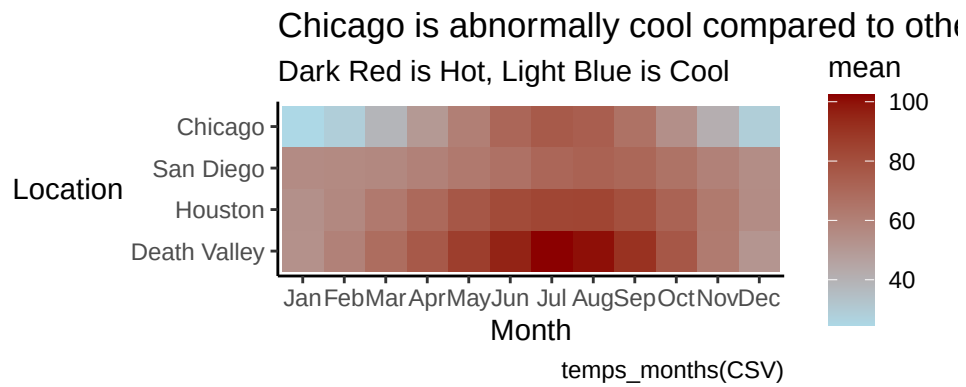


```
gf_tile( location ~ month , data = temps_months , fill = ~ mean) %>% gf_refine(coord_fixed( )) %>% gf_labs(x = "Month", y = "Location")
```

Chicago is abnormally cool compared to other locations



```
gf_tile(location ~ month, data = temps_months, fill = ~mean) %>% gf_refine(coord_fixed(), scale_fill_gradient2("mean", "low", "high"))
```



Blue is known for being colder while red is more associated with heat.